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Numerical and experimental analysis of heat and mass transfer in the drying process of the solar drying system

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ABSTRACT

Design in a solar food drying system is crucial for food drying performance. In order to make effective use of the sun, the use of air-heated solar collectors is gradually increasing in drying systems. Dryer room temperature and pressure distribution modelling is a very important factor for system design and product selection. In this study, the drying air speed, drying temperature, and product moisture content values of a solar drying system were numerically modelled in a time dependent manner. In numerical analysis, temperature and drying air velocity were defined as variable under initial and boundary conditions. Polynomial equations based on time obtained from drying room inlet temperature and drying air velocity data were used. The R^2 values of these equations were calculated as 0.96 and 0.91, respectively. COMSOL multiphysics program was used for temperature values, drying air velocity, product moisture content values and drying chamber pressure values modelling. Depending on the position of the sun and the time, the drying chamber inlet temperature values and air velocity values have been change. During the drying process of the product, moisture content values, heat and mass transfer values, drying efficiency values, diffusion coefficient and activation energy values were examined. Experimental data and computed fluid dynamics analysis (CFD) data were compared. Using the CFD numerical analysis program, the drying room temperature, moisture content and air velocity values were modelled with mean absolute percent error (MAPE) of 5.34%, 3.74% and 6.30%, respectively.

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1. Introduction

One of the best food storage methods today is drying. It causes the removal of moisture from a material with the heat and mass transfer mechanism [1]. In addition to products such as drying cereals, spices, seeds, broadcasting is used to reduce the moisture content of vegetables and fruits with a high moisture concentration [2]. Many valuable studies have been done in the field of food drying in the literature. Karacabey and Buzrul [3] have dried the pear product in a hot air dryer in their work. They described the pear drying data at 40, 50, 60, 70 and 80 °C with the Weibull model and observed that the shape parameter of the Weibull model is not temperature dependent. Kinetic analyses with published data have shown that the reduced Weibull model can be used successfully to identify the drying data of certain fruits. Lahsasni et al. [4] in their study, they conducted studies on the drying kinetics of pear fruit in

a convective solar dryer operating with an auxiliary heating system with adjustable air flow rate. In the drying temperature range of 50–60 °C, they carried out drying of the pear under the solar radiation of 250–900 W/m² at 23–34% relative humidity. Kaya et al. [5] Experimentally studied some thin layer drying properties of nettle and mint leaves in a convective dryer. They carried out drying experiments at 35, 45 and 55 °C air temperatures, 0.2, 0.4 and 0.6 m/s velocities and 40%, 55% and 70% relative humidity. They determined the activation energy values of the products dried using diffusion coefficient. Joardder et al. [6] carried out drying experiments in different operating conditions with pre-developed diffusion-based models to represent the drying behaviour of pears in a continuous convective dryer. They examined the chemistry, physical and thermal properties of pear fruit with time. They carried out drying experiments at a variable speed ranging from 0.5 to 1.5 m/s and with air containing 60% relative humidity at 30, 40 and 50 °C. They have shown that the drying models they use explain the drying behaviour of the pear product well enough. Hajar et al. [7] dried pears with a forced convection sun dryer. They have designed an air absorber for this sun dryer. They made the

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Nomenclature

A	Area (m ²)
c _i	The concentration of the species (mol/m ³)
C _p	Specific heat capacity at constant pressure (kJ/kgK)
D _{eff} , D _i	Effective moisture diffusivity (m ² /s)
D _o	Pre-exponential factor (m ² /s)
DR	Drying ratio
E _a	Activation energy (kJ/mol)
h _m	Mass transfer coefficient (m/s)
I	Total solar energy (kW)
I _c	Solar radiation on the collector (W/m ²)
J _i	The mass flux diffusive flux vector (mol/m ² s)
L	Thickness of the sample material (m)
m	Mass (kg)
m _a	Mass flow rate (kg/s)
M	Moisture content (kg/kg of dry basis)
MR	Moisture ratio
Q	Heat energy (kJ)
q	Heat flux by condition (W/m ²)
R	Universal gas constant (kJ/ mol K)
R _i	Reaction rate expression for the species (mol/m ³ s)
S	The second Piola-Kirchhoff stress tensor (Pa)
SAC	Solar air collector
SACD	Dryer with solar air collector
T	Temperature (°C)

U	Velocity vector (m/s)
V	Volume of the material (m ³)

Greek Symbols

η	Efficiency (%)
ρ	Density (kg/m ³)
α	Coefficient of thermal expansion (1/K)
τ	Viscous stress tensor (Pa)

Subscripts

a	air
c	collector
d	dry
db	dry basis
e	equilibrium
f	final
i	initial
m	material
o, out	outlet
r	solar radiation
t	anytime
trans	translational motion
w	water
wb	wet basis

absorbent plate from two corrugated aluminium sheet materials. They fixed these absorbent plates in the form of parallel cylinders along the collector. In the results obtained in this way, they demonstrated that the mass of the samples was reduced from 997.3 g to 135.13 g after 24 h of drying. They calculated the average thermal efficiency of the drying chamber as 11.11%.

Due to the development of computer systems in recent years, computerized analyses have been started in many fields. In drying systems, CFD analysis methods applications are gradually increasing for evaluating drying performance, analysing air speed and moisture distribution, commenting on heat and mass transfer values during drying and drying system designs. Studies related to CFD analysis in food drying are given in Table 1 in detail.

The common point of the 10 studies given in Table 1 is that their analyses were carried out using a numerical analysis program and they used time-independent models. Of these studies, only Kumar et al. [16] performed their analyses as time-dependent. In this study, for the first time in the literature, numerical analysis of time dependent parameters with the COMSOL program was performed. In the analyses, drying room temperature, air speed distribution and pressure effect were examined. In addition, product surface moisture changes were shown with the relevant contour values. The main purpose of this study is to show that the thermo-physical properties calculated with many data in drying processes in food drying systems can be determined more effectively with the help of numerical analysis program. This has an important role in determining the drying behaviour of the product to be dried in the designed system and can be used for many drying systems and products to be dried. At the same time, time-dependent numerical analyses of drying room temperature and air velocity values, which are generally accepted as constant in drying experiments (independent of time), were made. It is predicted that the findings can be used as preliminary information for drying system designs. The novelty of this study is that the parameters whose numerical analysis are made are modelled with the help of polynomial equations in a time dependent manner.

2. Material and methods**2.1. Experimental setup**

Drying test set consists of solar air collector (SAC), drying chamber and radial fan connected to air solar collector (Fig. 1). SAC used in experiments is 800mmx150mmx1400mm. SAC occurs (Fig. 2) transparent glass sheet, corrugated black matt painted corrugated stainless-steel sheet (thickness 0.5 mm) plate, plain black matt painted stainless steel sheet (thickness 0.5 mm) plate, stone wool (thickness 4 cm) insulating material and aluminium outer case (thickness 4 mm). On the other hand, SAC was directed south under the collector angle of 23.7° (local latitude 38.4°) and fixed with a stainless-steel leg at this angle. A centrifugal fan (0.0833 m³/s, 0.25 kW, 220 V, 50 Hz, 1380 rpm) provides conditioned air (speed, 0.4 m/s) in the air inlet portion of the SAC.

The drying process schematic representation and measurement points of the experiment set are shown in the Fig. 3. Temperature, air velocity, humidity measurements, radiation and product weight values were measured by sensors, anemometer, digital moisture meter, pyranometer and digital precision scales, respectively.

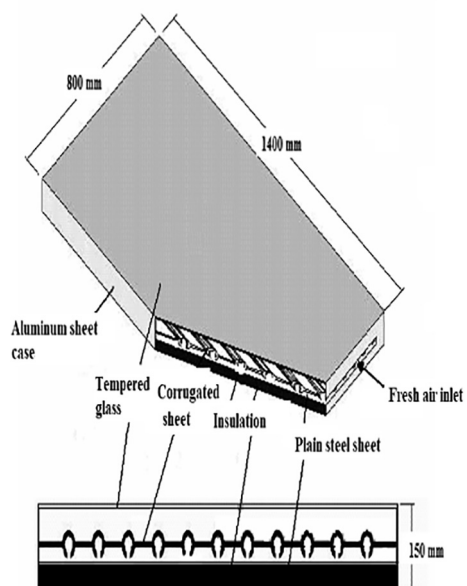
In drying experiments, Santa Maria type pear product was dried. Samples were sliced in a 14 mm thick semi-elliptical shape. Drying experiments were conducted Elazığ in Turkey under the open sun. In drying experiments, the ambient air temperature and relative humidity values, the inlet and outlet temperatures of the solar collector air, the humidity and the inlet–outlet temperatures of the air in the drying chamber, the wind speeds, the solar radiation values and the mass loss values of the pears dried in the drying chamber were measured with an interval of 30 min.

Drying chamber is made of aluminium material (thickness 2 mm) and designed in rectangular dimensions. (100 cm × 50 cm × 100 cm). A bent aluminium tube was used to transfer the heated air between the air solar collector and the drying chamber. The air inlet part of the drying chamber is manufactured with a hood to transmit hot air from the collector to the

Table 1

Some CFD analysis studies related to drying in the literature.

Ref.	Dryer System	Modelled Part	Model	Sample	Geometry	Parameters in The Model	Mesh Count& Element Number	Results
[8]	Infrared dryer	Drying Chamber	Steady State	Rice kernel	2D	Moisture content	19.064	The infrared radiation CFD model accurately estimated the average temperature and MC of the rice core.
[9]	Tray dryer	Horizontal plane in the tray	Steady State	Almond	2D-3D	Velocity, Moisture content, mass fraction of water vapour	4.5 M	Almond product developed CFD model for drying characteristic. As a result of CFD model, acceptance of air flow rate and heat input as constant for the unit weight of almond products indicated that the success rate of the model increased.
[10]	Microwave convective dryer	Drying Chamber	Steady State	Food	3D	Velocity, Pressure, Heat and mass transfer coefficient	460.000	Throughout the sample, it has been found to significantly affect drying kinetics due to variable heat and mass transfer coefficients. Due to air speed changes, a spatial variation of the heat transfer coefficient the mass transfer coefficients was observed at different limits of the sample.
[11]	Spray Dryer	Spray injector	Steady State	Droplet	2D	Velocity	35.600	The applied CFD model gave detailed information about the variation of droplet hydrodynamics and temperature over time. This information determined the time and place when the droplets dried.
[12]	Heat Pump Dryer	Drying Chamber	Steady State	Banana	2D	Velocity, moisture and temperature distribution	478.079	The temperature, moisture and speed of the air in the dryer and on the sample surface are modelled with CFD. The temperature values obtained by simulation showed the same decreasing tendency as the experimental temperature values of the heat pump dryer examined.
[13]	Solar + Infrared dryer	Solar air collector	Steady State	Melon Slices	3D	Velocity and Temperature distribution	2.01 M	The CFD analysis of solar air collector and drying chamber was done. As a result of the CFD analysis, the components of the drying system were described in detail and homogeneous and efficient drying was provided by proportional control of the product surface temperature.
[14]	Infrared dryer	Drying Chamber	Steady State	Apricot	3D	The heat and mass transfer characteristics	2.1 M	It was observed that the air flow rate variation played a stronger role on the cooling effect and evaporation rate compared to the temperature change.
[15]	Tray dryer	Product Surface	Steady State	Pineapple	2D	Temperature, Velocity, moisture content	2485	According to the results of Numerical Analysis, they stated that the best drying temperature for the pineapple product was 60 °C.
[16]	Convective dryer	Product Surface	Time dependent	Apple	2D	Temperature, Velocity, moisture content	3376	They stated that product drying kinetics can be calculated with numerical analysis and can be used in product quality estimation.
[17]	Convective	Product Surface	Steady State	Apple	2D	The heat and mass transfer characteristics	12,000	They obtained a theoretical model of shrinkage deformation in food drying.

**Fig. 1.** Solar air collector drying system.**Fig. 2.** Solar air collector detail.

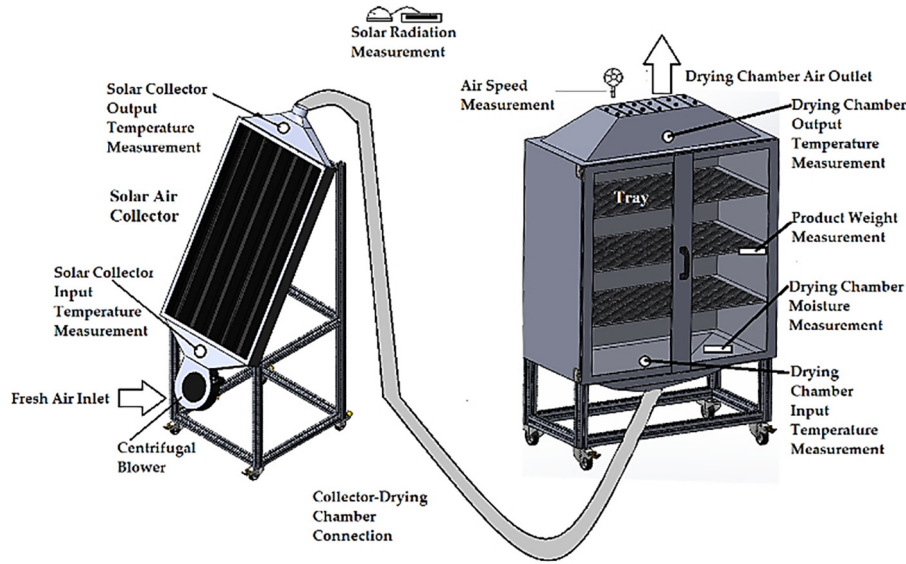


Fig. 3. The schematic representation of the experimental setup.

chamber. The air in the drying chamber exits to the atmosphere with the help of the opening and closing grille from the upper part of the chamber. There are 3 trays (90 cm × 40 cm) used for drying food in the drying chamber. The sensitivity values of the instruments and instruments used for the measurements performed in the experiments are shown in Table 2.

The uncertainty value of the measurements made in the experiment set was based on the method found by Kline and McClintock (Eq. (1)) [18,19]. In Eq. (1), x represents uncertainty properties and W represents uncertainty value. In this study, the uncertainty analysis calculated for various measurements is shown in Table 3.

$$W_x = \left[(x_1)^2 + (x_2)^2 + (x_3)^2 + \dots + (x_n)^2 \right]^{1/2} \quad (1)$$

2.2. System analysis

Some general equations used in the drying analysis of the drying system are given below. For the dry basis moisture content (M_{db}) and the wet basis moisture content (M_{wb}) values, the Eq. (2) and the Eq. (3) were used, respectively [20].

$$Mdb = \frac{W_i - W_f}{W_f} \quad (2)$$

$$Mwb = \frac{W_i - W_f}{W_i} \times 100 \quad (3)$$

Table 2
Measuring instruments and their features.

Parameter	Measurement Tool	Model	Precision
Moisture	Thermo-hygrometer	EXTECH, 444731, Chine	±0.1 °C
Temperature	J-type iron-constantan thermocouples	Elimko 6400, Italy	±0.1 °C
Air velocity	Digital Anemometer	LUTRON, AM-4201, Taiwan	±0.1 m/s
Initial and final moisture	Moisture Analyser	Shimadzu MOC3, Japan	±0.001 g
Solar radiation	Solar Pyranometer	Kipp and Zonen, Italy	±0.1 Wm ⁻²
Weight	Digital Balance	BEL, Mark 3100, Italy	±0.01 g

Table 3
Uncertainty analysis results.

Measurements	Unit	Uncertainty Value
SAC temperature	°C	±0.583
Drying chamber temperature	°C	±0.386
Moisture	RH	±0.1
Solar radiation (fixed SAC)	W/m ²	±0.506
Solar radiation (movable SAC)	W/m ²	±0.82
Product weight	gram	±0.5
Air velocity	m/s	±0.14
LDR sensor solar tracking	second	±0.05–0.1

In the Eqs. (2) and (3); W_i is wet weight and W_f is dry weight. Dimensionless humidity (MR) values were calculated using the Eq. (4) [21].

$$MR = \frac{M_t - M_e}{M_o - M_e} \quad (4)$$

$$W_x = \left[(x_1)^2 + (x_2)^2 + (x_3)^2 + \dots + (x_n)^2 \right]^{1/2}$$

In the Eq. (4); M_t is the moisture at the moment t , M_e is the equilibrium humidity and M_o is the initial humidity values.

2.3. Determination of the diffusion coefficient and the activation energy

In the food drying process, the movement of product moisture from the product to the product surface under certain drying conditions can be expressed as an effective moisture coefficient (D_{eff}). D_{eff} coefficient can actually be defined as a function of product moisture content value and product drying time. D_{eff} varies in proportion to the drying temperature. Fick's second law of diffusion is used to determine the D_{eff} coefficient [22]. The analytical solution of the second law developed by Fick and Crank [23] is expressed with the Eq. (5).

$$MR = \frac{8}{\pi^2} \exp\left(-\frac{\pi^2 D_{eff} t}{4L^2}\right) \quad (5)$$

In the Eq. (5), D_{eff} is the effective moisture diffusivity in m^2/s , t is the time (s), n is a positive integer, L and r are the half-thickness and radius of samples (m), respectively. For long drying periods, the Eq. (6) can be further simplified to only the first term of the series. The Eq. (6) is written in a logarithmic form as follows [24,25]:

$$\ln(MR) = \ln\left(\frac{8}{\pi^2}\right) - \left(\frac{\pi^2 D_{eff} t}{4L^2}\right) \quad (6)$$

Effective moisture diffusion varies depending on temperature. Arrhenius equation showing the change of effective moisture diffusion with temperature is given in the Eq. (7) [24,25]:

$$D_{eff} = D_0 \exp\left(-\frac{E_a}{R(T + 273.15)}\right) \quad (7)$$

Here D_0 is the pre-exponential factor of Arrhenius equation in m^2/s , E_a is the activation energy in kJ/mol , R is the universal gas constant in $kJ/molK$, and T is temperature in $^{\circ}C$.

2.4. Determination of convective mass transfer coefficients (h_m)

An effective parameter used to express drying behaviour, h_m is the surface properties of the materials to be estimated as it provides a good understanding of drying physics. Using MR , h_m values of products are calculated with the Eq. (8) [4].

$$h_m = \frac{V}{A_m t} \ln(MR) \quad (8)$$

In the Eq. (8), V is the volume of the material, A_m is the surface area of the material. The shape of pear slices is cylindrical.

2.5. Determination of drying and solar collector efficiencies

The drying air, which is heated by using the temperature of the absorber plates of the SAC, which is heated by the radiation from the sun, turns moisture away from the material and turns into useful heat. The heat generated by the SAC contains useful energy to evaporate moisture from the material.

The thermal efficiency of the SAC can vary depending on its geographical location, ambient air speed, ambient temperature, ambient humidity and many other factors. The efficiency of the SAC is calculated by the Eq. (9) [26].

$$\eta_{SAC} = \frac{Q_a}{I} = \frac{\dot{m}_a C_{pa} (T_{out} - T_i)}{A_c I_c} \quad (9)$$

In the Eq. (9), A_c is area of collector, I_c is solar intensity falling on the collector (W/m^2).

The Eq. (10) was used to calculate the drying efficiency of both drying systems at a constant fan speed. The drying efficiency (η_{SACD}) of the SACD's is estimated using,

$$\eta_{SACD} = \frac{(m_i - m_f) C_{pw} (T_{do} - T_{da})}{Q_a} \times 100\% \quad (10)$$

Subscripts, w is mentions water, do is outlet of drying chamber [20].

2.6. Determination of convective heat transfer coefficients (h_c)

In the present experimental study, the convective heat transfer coefficient was investigated for the convection flow on the product surface (flat plate). In the convective heat transfer calculations, it is assumed that the product is always at a constant temperature (T_s), and in convective drying, the product length (L) is too short to cause a turbulent flow.

The mean convective heat transfer coefficient values calculated for the laminar flow were calculated using the Pohlhausen equation (Eq. (11)). The Pohlhausen Equation and other correlations are expressed in the Eqs. (11)–(14) [27].

$$Nu_{lam} = 0.664 Re^{1/2} Pr^{1/3} \text{ (valid for } Re < 2 \times 10^5) \quad (11)$$

$$Nu = \frac{h_c L}{K_v} \quad (12)$$

$$Re = \frac{L V \rho_v}{\mu_v} \quad (13)$$

$$Pr = \frac{\mu_v C_v}{K_v} \quad (14)$$

The Re , Pr , and Nu numbers used in the drying process to calculate the heat transfer coefficient of the product were calculated as a result of the different physical properties of the humid air determined by the polynomial equations in the Eq. (15)–(18) [27,28]. T_i temperature is the temperature value obtained as a result of the average of T_e and T_s temperatures [29].

$$\rho_v = \frac{353.44}{(T_i + 273.15)} \quad (15)$$

$$K_v = 0.0244 + 0.6773 \times 10^{-4} T_i \quad (16)$$

$$C_v = 999.2 + 0.1434 T_i + 1.101 \times 10^{-4} T_i^2 - 6.7581 \times 10^{-8} T_i^3 \quad (17)$$

$$\mu_v = 1.718 \times 10^{-5} + 4.620 \times 10^{-8} T_i \quad (18)$$

2.7. Numerical analysis

The numerical analyses for the present experimental study were performed with COMSOL multiphysics 5.3a software. The numerical analysis is modelled as 2-D. The skewness of the analysis network structure is 0.85. The 2-D model structure of the mesh type is given in the Fig. 4. The solar assisted drying system designed here is basically the same as the convective dryer. Numerical modelling was performed for one product on a tray of the convective dryer. Therefore, the analysis was carried out under convective drying conditions. The model of the geometry analysed for this analysis is given in the Fig. 4. The drying chamber, product location and sensor positions (mm) used for analysis are shown in the Fig. 5.

The examined part of the drying chamber has a length of 1000 mm and a height of 150 mm. The food sample, assumed to be a semi-elliptical with a thickness of 14 mm and a radius of 30 mm, was placed 500 mm from the dryer inlet. Various convective desiccant analyses were performed similarly in the literature [30]. For analysis, measurements were taken from both sides of the product, at the geometric centre and at a distance equal to the geometric centre. The parameters on only one sample were examined in the drying chamber [11,31]. These measurements were then averaged. The difference of this study from the similar ones is the investigation of drying parameters depending on time and modelling of changes according to time. Chauhan et al. [32]



Fig. 4. Dryer chamber 2-D model.

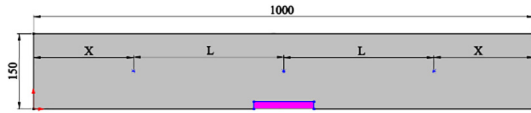


Fig. 5. Drying room and pear product 2-D view.

elaborated on time-dependent models in compilation studies. The time interval of the solution is $t = 60$ s. The simulation implementation was carried out on an Intel (R) i7 CPU-4700MQ @MAX 3.4 GHz, 32 GB installed RAM, running under Windows 8 (64 bit).

A drying simulation would be a “multiphysics” problem that combines fluid dynamics, mass transport, and, heat transfer. The multiphysics model can maximize physical understanding and calculation power by describing rightly all suitable phenomena. In this study, multiphysics interfaces were used to analyse heat transfer in airflow, mass transfer between air and product, and heat transfer between air and product.

The starting product temperature is $T_0 = 4$ °C. The mathematical expressions showing the changes in the drying room temperature parameter (T), drying room air speed parameter (V) and the wet base moisture content parameter (MC%) of the product in the drying room according to time (t) are given below in the Eq. (19) and the Eq. (20), respectively. Numerical analysis for the drying room was made according to these equations.

$$T = -1e - 20t^5 + 1e - 15t^4 - 4e - 11t^3 + 5e - 07t^2 - 0.0008t + 50.357 \quad (19)$$

$$V = 2e - 13t^5 - 3e - 10t^4 + 6e - 08t^3 + 4e - 06t^2 + 0.0016t + 1.1804 \quad (20)$$

The right boundary and initial conditions for the analysis have known to be pivotal in order to obtain a true simulation.

T_{air} = variable of the Eq. (19)

$T_{\text{product}} = 4$ °C

In the case of Laminar Flow interface, the following conditions are considered:

- “Wall” type condition: $u = 0$ for upper and lower boundary of the channel, with “no slip” specification for the walls. The upper and lower boundary have insulation boundaries conditions.
- “Inlet” type condition for channel inlet boundary
- U_{in} = variable of the Eq. (20)
- “Outlet” type condition for channel outlet boundary,
- with $P_0 = 0$
- The simulation was time dependent.
- The solution was performed the direct PARDISO solver.

The simulation equations (partial differential equations for numerical model) describing coupled heat and mass transfer in convective drying of the product were solved using the finite element method (FEM), COMSOL Multiphysics version 5.3a. In this analysis were used the default discretization for Laminar Flow is P1 + P1 elements. P1 + P1 elements – that is, piecewise linear interpolation for velocity and pressure [33]. This is suitable for most flow problems. The concentration variables were set to use linear elements by default parameter for the transport of diluted species physics. Heat transfer in fluids discretization was used linear but heat transfer in solids was used quadratic Lagrange [33].

This analysis model was created for a single section of the drying cabinet. There is airflow in the first part of the drying cabinet. Airflow and heat transfer analysis was carried out with momentum equations and heat transfer in fluids equations (Eq. (20), (21), (22)). The heat transfer from the air to the product has been solved with

the heat transfer in solids equation (Eq. 27). Transport diluted species equation (Eq. (24)) were used for moisture transfer from product boundaries. In the CFD analysis, modelling was carried out by considering the movement of the moisture (c) and the water mass balance in the drying air depending on the Fick diffusion law (Eq. (24)) [33–36]

$$\rho \frac{\partial u}{\partial t} + \rho(u \cdot \nabla)u = \nabla \cdot -pI + \mu(\nabla u + (\nabla u)^T) + F \quad (21)$$

$$\rho \nabla \cdot (u) = 0 \quad (22)$$

$$\rho C_p \left(\frac{\partial T}{\partial t} + u \cdot \nabla T \right) + \nabla \cdot (q + q_r) = -\alpha_p T \left(\frac{dP}{dt} + u \cdot \nabla p \right) + \tau : \nabla u + Q \quad (23)$$

$$\frac{\partial c_i}{\partial t} + \nabla \cdot J_i + u \cdot \nabla c_i = R_i \quad (24)$$

In the Eq. (24) J_i is the mass flux diffusive flux, u is the mass averaged velocity vector, R_i is a reaction rate expression for the species, D_i denotes the diffusion coefficient and, c_i is the concentration of the species.

Heat Transfer in Fluids (Eq. (22)) and Solids (Eq. (25)) equations used in this study are both as follows. [33].

$$\rho C_p \left(\frac{\partial T}{\partial t} + u_{\text{trans}} \cdot \nabla T \right) + \nabla \cdot (q + q_r) = -\alpha T : \frac{dS}{dt} + Q \quad (25)$$

Continuity Equation, Momentum Equations (Eq. (21), (22)), and Diluted Species Transport Equations (Eq. (24)) used in this analysis are as below. In the Eq. (21), (22), ρ is the density in air, t is the time (s), u is the velocity, p is the pressure (Pa) μ is dynamic viscosity of the air, T is the air temperature (K), and F is the sum of all forces acting on the particle. In the Eq. (23), (25) ρ is the density air or product, u is the velocity, t is the time, T is the air or product temperature (K), q is the heat flux by conduction, q_r is the heat flux by radiation, C_p is the specific heat capacity at constant stress, Q contains additional heat sources, α is the coefficient of thermal expansion, S is the second Piola-Kirchhoff stress tensor, u_{trans} is the velocity vector of translational motion.

MAPE error analysis was used to determine the accuracy of CFD analysis data according to experimental data. The MAPE error analysis is calculated with the help of the Eq. (26) [37,38].

$$MAPE = \frac{\sum \frac{|A-F|}{A} \times 100}{N} \quad (26)$$

In the Eq. (26), A is actual value, F is forecast value, N is number of observations, and the vertical bars stand for absolute values.

The consistency of results in CFD analysis is known to depend on the solution mesh structure. Therefore, it should be demonstrated in all studies that the result is independent of the solution mesh structure. In this study, three different solution mesh structures in the Table 4 were examined and it was seen that the solution obtained was independent of the mesh structure. The best results were obtained in different meshes for temperature and the results are shown in the Fig. 6. According to the Fig. 6, the best result for the drying chamber inlet temperature values was obtained by using mesh 1 in different mesh structures. The Fig. 7

Table 4
The mesh independency.

Mesh	Number of Elements	Max. Skewness	Average Quality
1	70,745	0.8523	0.95264
2	38,944	0.8507	0.94781
3	26,002	0.8475	0.94155

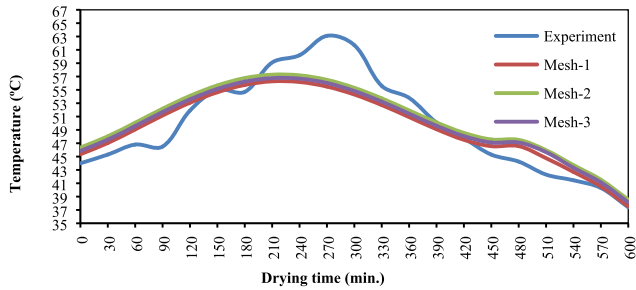


Fig. 6. The mesh independency for dryer chamber average temperature.

shows an example of the mesh model prepared for this numerical study.

3. Result and discussion

Moisture content, heat and mass transfer, activation energy and diffusion coefficient calculations were made for the product dried in the drying system. The drying efficiency and solar air collector efficiency calculations of the convective drying system with solar air collector used for drying pears were calculated. In addition, numerical analysis of drying room air speed change, drying room temperature change and wet base moisture changes of the drying room product were performed depending on the time. The findings are shown in more detail by the following figures.

The Fig. 8 shows the change of dry base moisture content values of pear slices with time. The moisture content was determined to be in an exponential form, which is the general characteristic of the drying curves of the curve obtained when the experiment data was drawn against time. The initial moisture content of the product was $4.91 (g_{\text{water}}/g_{\text{drymatter}})$, and the drying process was terminated at a value of $0.27191 (g_{\text{water}}/g_{\text{drymatter}})$, where the moisture content remained constant. This value was used as the equilibrium moisture content in later calculations. Time to reach equilibrium moisture in the SADC was determined as 600 min.

For the normalization of drying curves created by experimental findings, the dry base moisture content values that changed over time were converted into a dimensionless parameter, which is also called time-varying humidity. In Fig. 9, it was shown that moisture content values of pear samples decreased from 1 to 0.000062 in drying process in the SADC.

The change of the thermal efficiency values of the SAC during the drying time is shown in the Fig. 10. It can be observed that the thermal efficiency of the solar air collector changes according

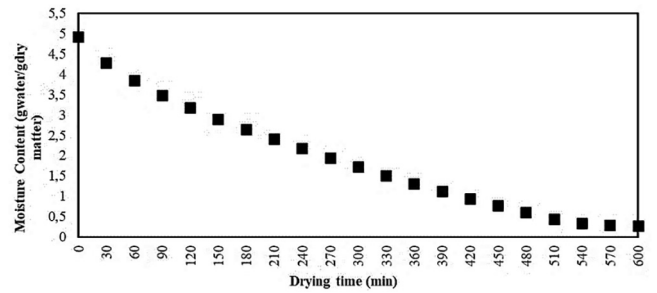


Fig. 8. The variation of dry base moisture content values with time in the SADC.

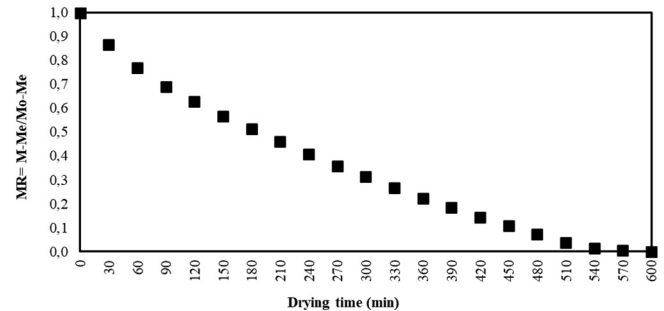


Fig. 9. The variation of moisture ratio values with time in the SADC.

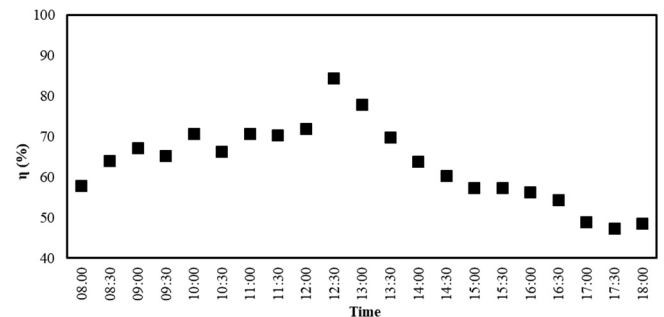


Fig. 10. The variation of collector efficiency values with time in the SAC.

to the time when the sun's rays are perpendicular to the location where the experiments are carried out.

In Fig. 11, the variation of the drying efficiency values of pear slices dried in the SADC with time are shown. During the first drying phase, the amount of excess liquid starts to move away from the product and then gradually decrease.

The change in the mass transfer coefficient values of pear slices dried in the SADC is shown in the Fig. 12. In the drying system, the mass transfer coefficient increased with the progress of drying

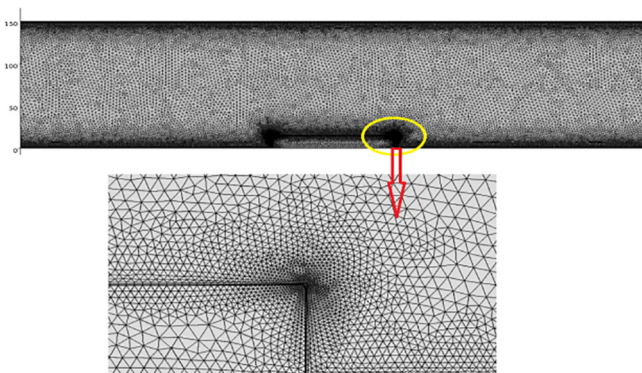


Fig. 7. Analysis mesh structure.

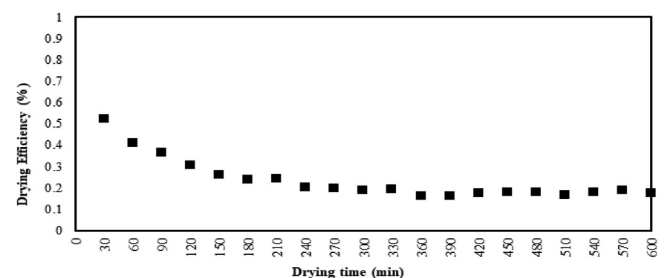


Fig. 11. The variation of drying efficiency values with time in the SAC.

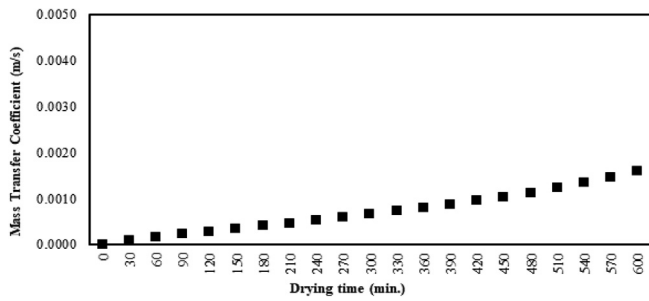


Fig. 12. Change of the h_m values according to time in the SACD.

time. It is shown in the Fig. 12 that the mass transfer increases during the product drying period and the average mass transfer value is 0.007 m/s.

The Fig. 13 shows the heat transfer values during the drying time of the product in SACD. Heat transfer coefficient values vary between 16.2–20.9 W/m²K. However, in this numerical study used heat transfer coefficient is constant 18.55 W/m²K.

In drying experiments, the activation energy values decreased in direct proportion with the drying efficiency. The E_a values calculated with experiment data are shown in the Fig. 14. According to the Fig. 14, the average E_a value was calculated as 58.4 kJ/mol.

Diffusion coefficient values increased in direct proportion with the mass transfer value. The D_{eff} values calculated with experiment data are shown in Fig. 15. According to the Fig. 15, the average D_{eff} value was calculated as 0.035×10^{-8} m²/s.

In Figs. 16, 17 and 18, changes in drying room air velocity, drying room temperature and product wet base moisture content values depending on time are shown. According to Figs. 16–18, it can be observed that the moisture content values, drying chamber temperature and drying chamber air velocity values of the pear sample are almost the same in experimental and numerical analysis before the drying process begins and during the drying phase. In the Fig. 19, the error values of the parameters that are numerically modelled during the drying period are given. The least error change occurred in moisture content analysis. The most errors occurred in drying air speed analysis. The error values (MAPE) of CFD for the moisture content, air velocity and temperature are 3.74, 6.30 and 5.34, respectively.

In this experimental study, in which time-based numerical analysis of parameters in the solar drying system was performed, product drying and drying cabinet models were successfully performed under varying temperature and air speed conditions during drying. A successful model has been established for new drying systems to be designed in the future. The compatibility of the numerical models with the experiments has been shown.

2D numerical analysis results of drying chamber pressure and drying air velocity values are given in the Fig. 20. The pressure and drying air velocity distribution contours within the drying

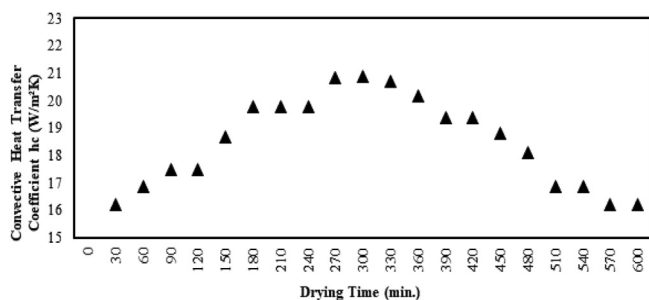


Fig. 13. Change of the h_c values according to time in the SACD.

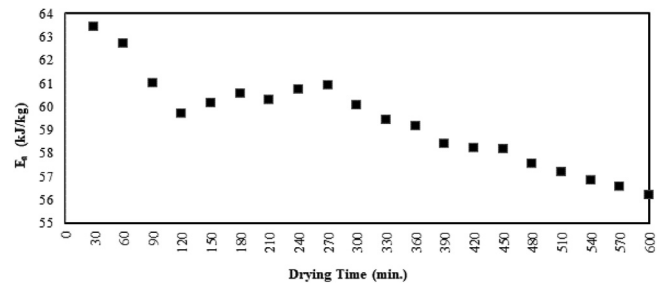


Fig. 14. The variation of activation energy values according to the drying time in the SACD.

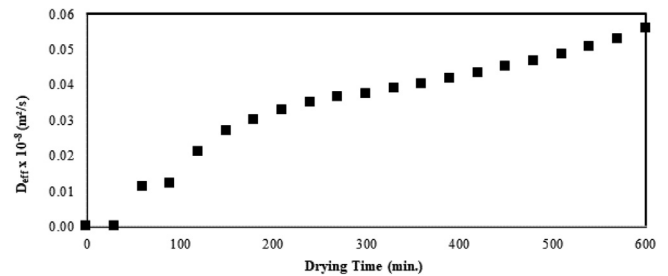


Fig. 15. The variation of diffusion coefficient values according to the drying time in the SACD.

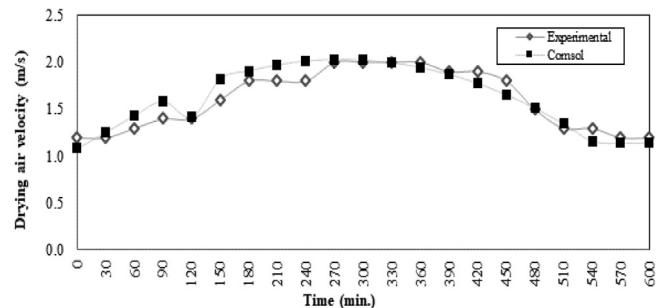


Fig. 16. The variation of drying air velocity experimental and numerical analysis values according to the drying time.

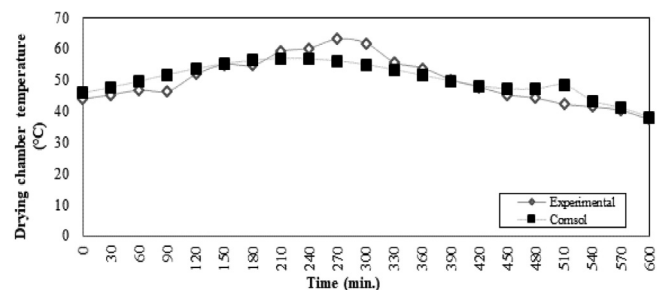


Fig. 17. The variation of drying chamber temperature experimental and numerical analysis values according to the drying time.

chamber are shown as A and B, respectively. Colour scales are located right next to these contours. As in the experimental study, air velocity and pressure are defined in numerical study. As a result, the air velocity and pressure change over time in the drying chamber. When it is examined at the beginning, the sharp edge of the product causes the air to change direction at the point where it hits first and an increased air movement just above the product.

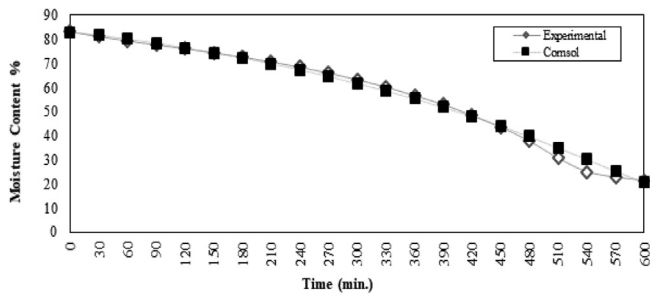


Fig. 18. The variation of moisture content experimental and numerical analysis values according to the drying time.

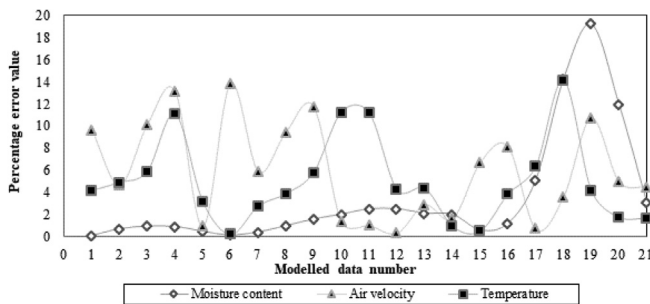


Fig. 19. Error values resulting from numerical analysis of parameters.

This situation clearly shows the pressure and speed distribution contours. This shows that when more than one product is placed in the drying tray, the distance between the products (horizontal and vertical) will be important. It is seen that the air velocity is zero at the point where the air flow hits the product. The pressure on the product is lower than the other parts of the drying chamber, which is considered to be due to the increase in the speed of the air flow. The air velocity has increased somewhat due to the cross section narrowing in the upper part of the region where the product is located. When a drying chamber with such dimensions designed, the air speed changes are low enough to ignore the pressure drop at the speeds of 0–2.5 m/s.

Product moisture distribution and temperature contours for three different times are given in the Fig. 21. In order to better visualize the moisture distribution in the product, the screen image was enlarged by zooming in. Product moisture distribution takes place more than the edges of the product. Less moisture transfer occurs from the top surface of the product. The reason for this is that the point where the drying air first touches the edge of the product. When the temperature distribution inside the cabin is examined, it is seen that the temperature around the product decreases depending on the heat and mass transfer from the product. Due to the direction of the flow movement, it is seen that the fluid temperature at the back of the product is lower than other regions. The results of the analysis were calculated using the average absolute percentage error (MAPE) calculation method and the error rate between the experimental data. The product moisture

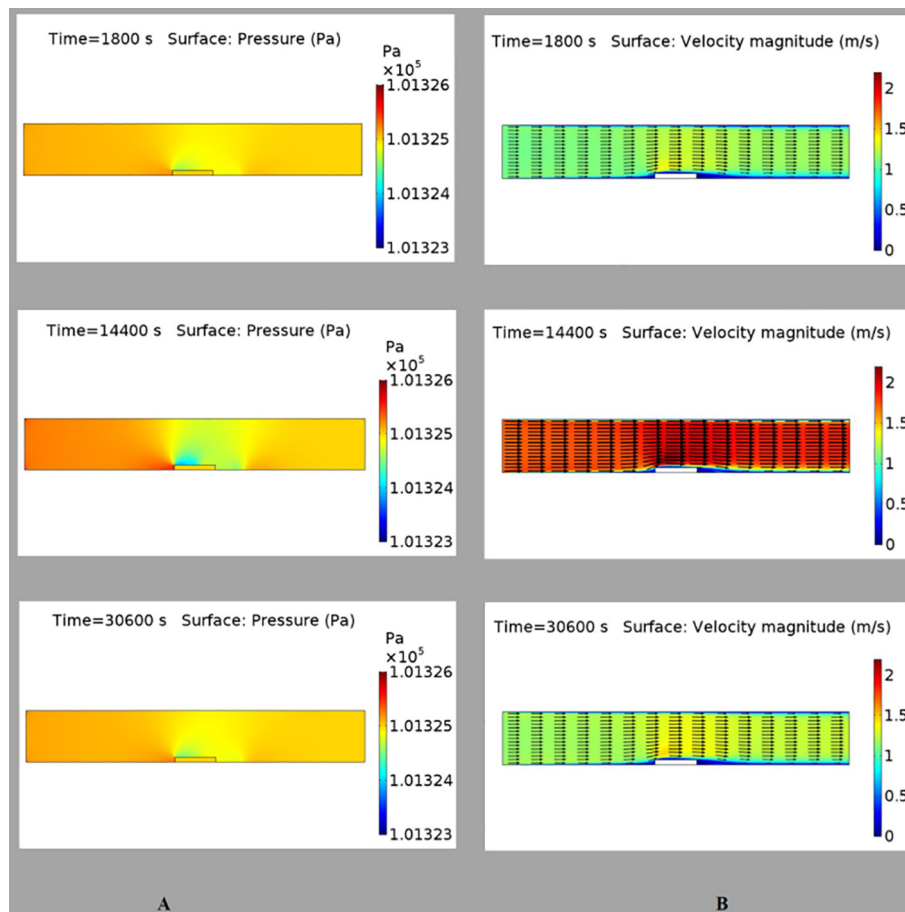


Fig. 20. Change of 2D numerical analysis results of pressure and drying air velocity values over time.

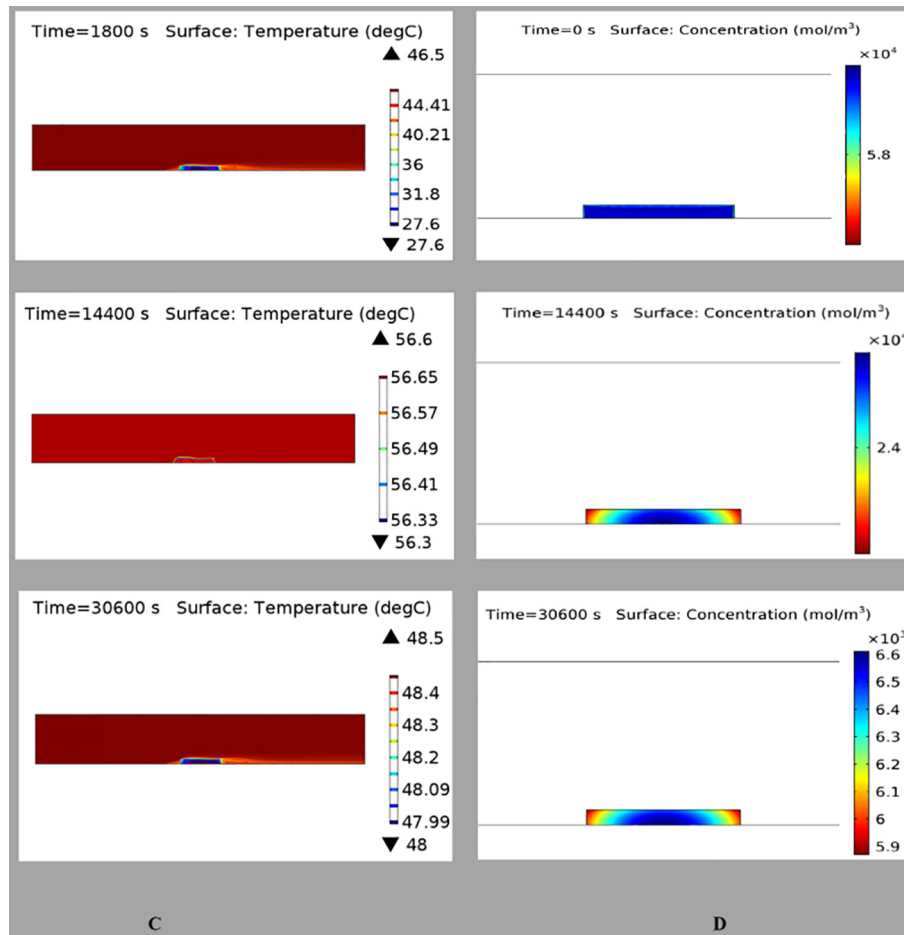


Fig. 21. Change of 2D numerical analysis results of temperature and moisture content values over time.

content is modelled with an error of 3.74%. With the COMSOL 5.3 a, the in-product moisture distribution based on time has been successfully modelled. In these dryers to be designed with time-based analysis, the opportunity to determine the optimum drying conditions of the product was offered.

Most of the numerical modelling studies related to drying in the literature have been done for time-independent parameters [8–15,39]. The difference of this study is that the parameters examined in numerical analysis were modelled in a time dependent manner.

4. Conclusions

In this study, a detailed examination of the drying process carried out in a solar food drying system was carried out. The parameters of heat and mass transfer, activation energy, diffusion coefficient and moisture content, which can be examined by a drying process, were calculated. At the same time, time-dependent numerical analyses were made for the drying chamber pressure, temperature, air velocity and product moisture content values in the drying system. Thanks to the 2D contours obtained as a result of the analysis, the numerical model was more clearly expressed. The values found in the study are given below as items.

- 1- During the drying period, the moisture basis (dry basis) values of the product decreased from 4.91 ($\text{g}_{\text{water}}/\text{g}_{\text{dry matter}}$) to 0.273 ($\text{g}_{\text{water}}/\text{g}_{\text{dry matter}}$). The average dry base moisture content value was found to be 2 ($\text{g}_{\text{water}}/\text{g}_{\text{dry matter}}$).

- 2- The D_{eff} values of the product increased during the drying period of the solar drying system. The average value of the diffusion coefficient, which is directly proportional to the mass transfer of the product in the drying system, was calculated as $0.035 \times 10^{-8} \text{ m}^2/\text{s}$.
- 3- In drying experiments, activation energy values decreased in direct proportion with drying efficiency values. The average E_a value calculated with the experimental data was calculated as 58.4 kJ/mol. The average drying efficiency value during drying is 29%. Due to the rapid decrease in product moisture rate between 0 and 210 min, activation energy and drying efficiency also decreased depending on humidity values.
- 4- In the drying process, it is calculated that the mass transfer increases during the drying period of the product and the average mass transfer value is 0.007 m/s. One of the most important factors affecting mass transfer is the diffusion coefficient of the product. As a result of the calculations, it can be said that the mass transfer and diffusion coefficient change in direct proportion.
- 5- Convective heat transfer values of the product in the drying process have changed depending on the temperature of the drying air heated by the collector. The heat transfer values have changed in parallel with the change in the collector efficiency values in Fig. 10. The average heat transfer value calculated with experimental data is $18.55 \text{ W/m}^2\text{K}$. The heat transfer value corresponding to the highest collector efficiency value (83.2%) is $20.9 \text{ W/m}^2\text{K}$.

6- Numerical models of drying cabinet and product drying were made time dependent. Drying cabin inlet temperature and air velocity, which change according to the position of the sun, were defined as variable. Contours of the speed and pressure distribution inside the drying cabinet were shown. In-product moisture distribution contours are shown. The compatibility of this model with experiments has been demonstrated. The CFD drying room temperature value, humidity content value, and velocity values (mean absolute relative error) were modelled with an error rate of 5.34%, 3.74% and 6.30%, respectively.

With the present study, time dependent equations were produced for temperature and air velocity parameters, which are generally regarded as time independent and constant, and these parameters were numerically modelled. In this way, the effect of temperature and speed parameters on the product during the drying process can be examined better. The methods and models used can enable many researchers who conduct experimental studies on product drying to conduct more detailed drying experiments by examining the time dependent change of parameters.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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